MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGULATOR, NEGATIVE, POWER, KEY CODE ARN

This amendment forms a part of Military Specification MIL-M-28787/192B(EC), dated 18 February 1981, and is approved for use by the Naval Electronic Systems Command, Department of the Navy , and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 AND 5

* TABLE I, delete and substitute new table I as printed on pages 2 and 3 of this amendment.

PAGE 7

* 6.4, Load regulation: Delete "0.2%" and substitute "0.5%".

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* TABLE V, delete and substitute new table V as printed on page 4 of this amendment.

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* TABLE VII, Test condition F5, Min and Max columns: Delete "-10" and "10" and substitute "-15" and "15" respectively.

* TABLE VII, Test condition F6, Min and Max columns: Delete "-10" and "10" and substitute "-25" and "25" respectively.

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* FIGURE 9, add note 16 as follows:

"16. Do not apply item P01 power to the module until immediately before the reading is taken at each test condition and remove power immediately after the reading is taken to prevent the module from heating up."

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* TABLE VIII, Max column: Delete "125" and substitute "129".

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FIGURE 15, NOTE 1: Delete "$t_{THL} = t_{TLH} = 10 \pm 5$ ns and substitute "$t_{THL} = t_{TLH} = 1.0 \pm 0.1$ µs".

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Agent:
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(Project 5963-N453)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	Isolation	II	X	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
B5	Continuity	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
C1	Output voltage program	IV	X	-5.05	-4.95	-5.10	-4.90	-5.10	-4.90	-5.15	-4.85	Vdc
C2	Output voltage program	IV	X	-15.30	-14.70	-15.45	-14.55	-15.45	-14.55	-15.60	-14.40	Vdc
D1	DC input voltage regulation	V	X	-5.05	-4.95	-5.10	-4.90	-5.10	-4.90	-5.15	-4.85	Vdc
D2-D6	DC input voltage regulation	V	X	← ±23 mV variance max from D1 →								
D7	DC input voltage regulation	V	X	← ±25 mV variance max from D1 →								
D8	DC input voltage regulation	V	X	-15.30	-14.70	-15.45	-14.55	[illegible]	-14.55	-15.60	-14.40	Vdc
D9-D12	DC input voltage regulation	V	X	← ±45 mV variance max from D8 →								
E1, E2	Input modulation regulation	VI	X	0.1	750.0	0.1	750.0	0.1	750.0	0.1	750.0	μVrms
E3, E4	Input modulation regulation	VI	X	0.01	2.25	0.01	2.25	0.01	2.25	0.01	2.25	mVrms
F1	Output load regulation	VII	X	-5.05	-4.95	-5.10	-4.90	-5.10	-4.90	-5.15	-4.85	Vdc
F2	Output load regulation	VII	X	← ±2.5 mV variance max from F1 →								
F3, F4	Output load regulation	VII	X	← ±10 mV variance max from F1 →								
F5	Output load regulation	VII	X	← ±15 mV variance max from F1 →								
F6	Output load regulation	VII	X	← ±25 mV variance max from F1 →								
F7	Output load regulation	VII	X	← ±50 mV variance max from F1 →								

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
F8	Output load regulation	VII	X	-15.30	-14.70	-15.45	-14.55	-15.45	-14.55	-15.60	-14.40	Vdc
F9	Output load regulation	VII	X	← ±7.5 mV variance max from F8 →								
F10-F13	Output load regulation	VII	X	← ±30 mV variance max from F8 →								
F14	Output load regulation	VII	X	← ±150 mV variance max from F8 →								
G1	Output current limiting	VIII	S	100	129	97	129	88	140	86	143	mAdc
G2	Output current limiting	VIII	S	200	245	194	252	176	274	172	279	mAdc
G3	Output current limiting	VIII	S	300	365	291	376	264	409	252	416	mAdc
G4	Output current limiting	VIII	S	400	485	388	500	352	543	344	552	mAdc
H1, H2	Output noise	IX	X	0.0	250.0	0.0	250.0	0.0	250.0	0.0	250.0	μVrms
J1, J2	Output impedance check	X	S	5	500	5	500	5	500	5	500	μVrms
J3	Output impedance check	X	Q	10	1,000	10	1,000	10	1,000	10	1,000	μVrms
K1	Output voltage latch	XI	S	0.650	0.950	0.650	0.950	0.500	0.950	0.500	0.950	Vdc
K2	Output voltage latch	XI	S	-5.05	-4.95	-5.10	-4.90	-5.10	-4.90	-5.15	-4.85	Vdc
L1	Load transient response	XII	S	0	150	0	150	0	150	0	150	mV
L2	Load transient response	XII	S	0	450	0	450	0	450	0	450	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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TABLE V. DC input voltage regulation test.

Test condition	Parameter	PO1 (V dc)	S1 position	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	DC input voltage regulation	10 ±0.10	1	8	-5.05	-4.95	Vdc	7
D2		15 ±0.15	1		← 1/ →			
D3		20 ±0.20	1					
D4		25 ±0.25	2					
D5		30 ±0.30						
D6		35 ±0.35						
D7		40 ±0.40			← 2/ →			
D8		20 ±0.20		7	-15.30	-14.70	Vdc	
D9		25 ±0.25			← 3/ →			
D10		30 ±0.30						
D11		35 ±0.35	3					
D12		40 ±0.40	3					

1/ V_{out} should not vary more than ±23 mV from value of D1.

2/ V_{out} should not vary more than ±25 mV from value of D1.

3/ V_{out} should not vary more than ±45 mV from value of D8.

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